

PREDICTIVE SKILL-SEAM ACTION MODELS FOR ROBOT SKILL COMPOSITION

Anonymous authors

Paper under double-blind review

ABSTRACT

Robot skills that work individually can fail when chained: the terminal state of one skill may fall outside the next skill’s attraction basin, cross a high-energy barrier, or enter a contact mode where descent is no longer smooth. We frame that handoff as a local world/action-modeling problem: a compact predictive interface between a skill library and a planner for handoff consequences. Given terminal-state evidence, next-skill basin/descent estimates, and a candidate transition, the interface estimates whether an action/skill composition will fail, diagnoses the likely reason, chooses accept, repair, probe, abstain, or alternate transition, and turns the observed result into planner-edge evidence so later plans can prefer reliable seams, repair uncertain ones, or avoid repeated failures. A barrier-certified energy composer implements that interface, acting on a skill edge only when basin overlap, barrier height, descent continuity, repair cost, and fixed-risk calibration are jointly favorable. In a frozen local rollout suite with 12 methods, 6 task families, 8 seam regimes, and 10 paired seeds, the composer reaches hard-slice success 0.80171 and utility 0.88827, compared with 0.71711 and 0.65310 for the strongest non-oracle predecessor. It reduces seam failure, barrier violation, damage, calibration error, and realized seam breach while improving basin alignment and descent continuity. The current evidence supports a bounded claim: treating a skill seam as a local world/action model improves composition when the relevant basin, barrier, and descent quantities are estimable and their outcomes update future planning. This is one local step toward adaptive physical world/action models, while external robot or high-fidelity validation remains necessary before deployment-level claims.

1 MOTIVATION

A reusable robot skill library still leaves a planning problem: the robot must know what the end of one skill makes possible for the next. This paper focuses on that seam, the narrow place where an action model can be smaller than a full simulator but richer than a graph edge. The model predicts what a completed skill makes true, risky, repairable, or impossible for the next skill. Its job is not to operate either primitive or replace the planner; it is to make the transition legible enough for planning to choose the next action. At this scale, the useful prediction is not a full future video, but whether a specific transition creates a feasible start state, a repairable state, an uncertain state worth probing, or a state the planner should avoid. Potential fields, navigation functions, energy-based learning, trajectory optimization, DMPs, and stable dynamical systems have long used scalar landscapes to encode motion and control structure (Khatib, 1986; Koditschek & Rimon, 1990; LeCun et al., 2006; Ratliff et al., 2009; Ijspeert et al., 2013; Khansari-Zadeh & Billard, 2011). Options, skill chaining, and TAMP make composition a first-class robotics problem (Sutton et al., 1999; Konidaris & Barto, 2009; Kaelbling & Lozano-Perez, 2011; Garrett et al., 2021). Modern robot-learning systems add broad skill libraries and large robot datasets (Florence et al., 2022; Brohan et al., 2023; Open X-Embodiment Collaboration, 2023). The gap targeted here is not low-level skill learning or a new manipulation controller; it is the missing action-conditioned physical interface between a skill library and a planner. Before execution, the robot should estimate whether a proposed edge will land inside the next skill’s feasible start set or whether it should repair, probe, abstain, or choose a different transition. After execution, it should keep the outcome as planning evidence: which handoff was reliable, which needed repair, which called for abstention, which should be avoided later, and which missing bridge skill would make future plans less brittle. The core failure mode is

simple: skill one ends successfully, but its terminal distribution lies near a ridge or outside the basin of skill two. A module graph may mark the edge legal, while execution requires a high-energy repair, crosses a barrier, or enters a contact mode where the next controller no longer descends. The paper asks whether an explicit seam model can make composition more reliable by predicting whether a proposed transition will fail before the handoff, diagnosing the likely reason, deciding when to accept, repair, probe, abstain, or choose a different transition, and using the outcome to improve later composition choices. Contact-rich examples matter here because they expose action consequences that a motion-only edge label misses; they are a testbed for the seam model, not the identity of the paper. The broader question is how a robot represents the physical consequences of a skill transition, notices when that representation is incomplete, and adapts future planning from the outcome. We use the world/action-model view at a deliberately local scale: the model is not a whole robot simulator, and the prediction-action-update loop is limited to the handoff. That restraint is intentional. The paper studies the small interface where a planner asks, before committing to the next skill, what this transition is likely to do, why it might fail, which response is warranted, and what should be remembered afterward. Concretely, the contribution is a seam-level predictive interface: estimate terminal/basin/barrier/descent/risk quantities, diagnose the likely failure mode, choose accept, repair, probe, abstain, or an alternate transition, and write the outcome back to planner-edge memory so future planning queries do not treat every graph edge as equally safe. The energy composer is one implementation of that interface; the paper’s claim is the smaller and testable one that skill seams need a local world/action model.

2 PROBLEM SETUP

Let skill i induce a terminal distribution $p_i(x_T)$ and skill j have an attraction basin $\mathcal{B}_j$ under local controller π_j . We treat the handoff as a small predictive model of action consequences at the seam. A composition edge $i \rightarrow j$ is acceptable only if terminal samples have sufficient basin overlap, low barrier height, and positive descent continuity. The action set at this boundary is deliberately planner-facing: accept the edge, repair the handoff, probe an uncertain seam, abstain, or route to an alternate transition. Operationally, the seam model maps a candidate transition to a consequence tuple: predicted basin compatibility, barrier exposure, descent continuity, repair cost, calibrated risk, and a failure label such as basin mismatch, high barrier, contact-mode discontinuity, model uncertainty, or missing bridge skill. We use world/action model in this limited sense: the output is not a general simulator and does not try to predict every future observation. The local world is the physical seam state around a handoff; the action is the proposed skill transition, repair, probe, abstention, or alternate edge. The model predicts how that action changes the next skill’s feasible start set, turns the prediction into a decision, and records the result as an edge belief for future plans. An abstention or repair recommendation is therefore not a bookkeeping flag; it is a physical statement about what the current skill library cannot yet compose reliably. More explicitly, the interface can be read as $M_\theta(i, j, h; \mathcal{D}_t) \mapsto (\hat{O}, \hat{B}, \hat{D}, \hat{C}, \hat{R}, \hat{y}, \Delta b)$, where h is a candidate handoff, $\mathcal{D}_t$ is the accumulated seam evidence, $\hat{y}$ is the diagnostic label, and Δb is the planner-edge belief update. The adaptation is modest but important: a failed or successful seam changes which transition the next planning query prefers, probes, repairs, or refuses, so the model improves future planning rather than only scoring one edge. We score candidate handoff h by

$$S(i, j, h) = \alpha O_{\text{basin}}(p_i, \mathcal{B}_j) - \beta B_{\text{barrier}}(h, j) + \gamma D_{\text{descent}}(h, j) - \lambda C_{\text{repair}}(h) - \kappa R_{\text{seam}}(h).$$

Here O_{basin} measures terminal/basin overlap, B_{barrier} measures energy barrier height, D_{descent} checks whether energy decreases after handoff, C_{repair} is seam-repair cost, and R_{seam} is calibrated seam risk. The proposed composer accepts the edge only when predicted seam risk is below a declared fixed-risk budget; otherwise the failed prediction becomes a planning signal for repair, diagnostic probing, abstention, or a different transition. Across attempts, the same interface can update a planner’s edge beliefs: which handoffs are reliable, which need a bridge action, which need more evidence, and which should be avoided until the skill library changes.

3 SKILL SEAMS AS PREDICTIVE INTERFACES

A sequence can be graph-valid while physically unsafe at the seam. If terminal states from the first skill do not overlap the next basin, the second skill starts outside its attraction region. If a high barrier

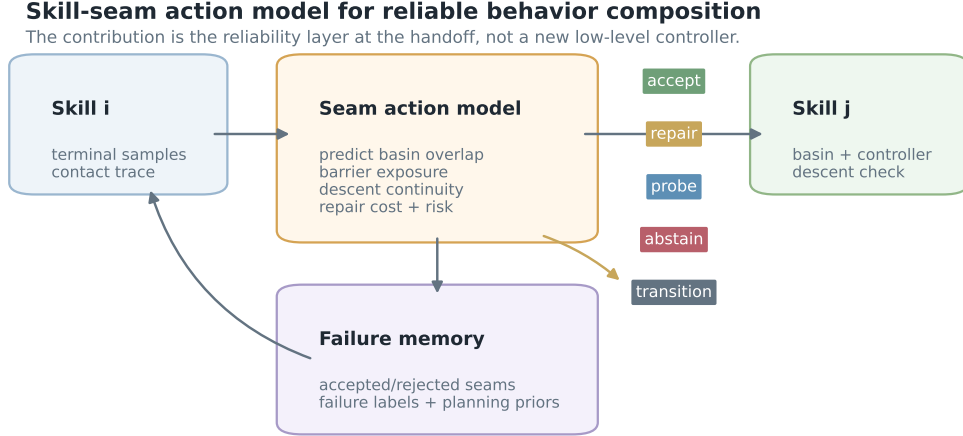


Figure 1: Skill-seam action interface. The method predicts local handoff consequences, diagnoses likely failure modes, and converts that evidence into accept, repair, probe, abstain, or transition decisions plus planner-edge updates for future planning.

separates the terminal set from the basin, a repair may be possible but costly or unsafe. If the contact mode changes, the energy function may no longer be conservative enough to certify descent. These are not merely controller details; they are action-conditioned predictions about what a skill transition will do to the physical state available to the next skill. The model output is therefore not just a compatibility number. It is a compact answer to four planning questions: will this handoff fail, why might it fail, should the system repair, probe, abstain, or route through another transition, and what should the planner learn for later plans? The important prediction is the one that changes the next action choice, including the choice not to execute. This is the paper’s intended world/action-model reading: prediction matters because it changes action selection and future planning memory. This gives a bounded claim. When basin overlap, barrier height, and descent continuity are identifiable enough to reject bad seams and preserve useful ones, a barrier-certified implementation can make skill composition more reliable. Its value is not only higher success, but more structured failure information: the planner can learn which transitions are reliable, which require repair or a diagnostic probe, and which should be avoided. What makes this a seam action model rather than only a score threshold is the full loop: predict the local consequence before execution, choose a transition-level response, observe the outcome, and update the edge memory that future plans query. It must fail or abstain when the low-level primitive is broken, the goal is semantic rather than physical, the landscape is miscalibrated, or real-time constraints prevent search.

4 EVALUATION PROTOCOL

The evaluation protocol is fixed before interpreting final results. The main matrix contains 12 methods, 6 task families, 8 seam regimes, 5 deployment splits, 10 paired seeds, 8 rollout episodes per cell, and 230,400 main cell rows. Ablations add 38,400 cells, stress sweeps add 161,280 cells, fixed-risk tests add 107,520 cells, and the failure audit contains 24 cases. The previous proposed method remains a named baseline. The protocol measures both composition outcomes and seam-interface behavior: predicted risk, diagnostic label, decision, realized breach, and planner-edge update.

5 MAIN RESULTS

The strongest non-oracle comparator is `proposed_energy_landscape_composer_v4_1`. The proposed composer improves hard success by 0.08460 and hard utility by 0.23517; paired hard-utility wins are 10/10. The oracle remains stronger with success 0.90967 and utility 1.13988, so the local problem is not solved. The intended readout is therefore not only success; it is whether a seam action model

gate	status
ablation_success_margin_ge_0.015	pass
ablation_utility_margin_ge_0.030	pass
barrier_violation_delta_le_-0.020	pass
basin_alignment_delta_ge_0.030	pass
composition_cost_delta_le_0	pass
damage_rate_delta_le_-0.005	pass
descent_continuity_delta_ge_0.030	pass
failure_cases_ge_24	pass
hard_success_margin_ge_0.030	pass
hard_utility_margin_ge_0.050	pass
paired_hard_utility_wins_ge_8	pass
realized_seam_breach_delta_le_-0.020	pass
risk_calibration_error_delta_le_-0.010	pass
seam_failure_delta_le_-0.020	pass
stress_endpoint_success_margin_ge_0.030	pass
strict_fixed_risk_breach_le_0.020	pass
strict_fixed_risk_coverage_ge_0.550	pass

Table 1: Local evidence gates used for the seam-composition audit. External robot or high-fidelity validation is evaluated separately.

method	succ.	utility	seam	barrier	basin	descent	damage	cost	breach
oracle_basin_composer	0.910 $\pm$ 0.005	1.140 $\pm$ 0.005	0.032	0.053	0.864	0.844	0.067	0.151	0.039
barrier_certified_energy_composer_v5	0.802 $\pm$ 0.007	0.888 $\pm$ 0.007	0.069	0.085	0.757	0.741	0.071	0.189	0.095
proposed_energy_landscape_composer_v4.1	0.717 $\pm$ 0.008	0.653 $\pm$ 0.008	0.118	0.126	0.677	0.663	0.076	0.235	0.171
stable_dmp_handoff	0.637 $\pm$ 0.008	0.391 $\pm$ 0.008	0.167	0.166	0.546	0.535	0.080	0.298	0.242
tamp_feasibility_screen	0.638 $\pm$ 0.008	0.361 $\pm$ 0.008	0.173	0.171	0.534	0.513	0.080	0.334	0.251
energy_compatibility_heuristic	0.620 $\pm$ 0.008	0.303 $\pm$ 0.008	0.193	0.190	0.516	0.503	0.083	0.305	0.284
residual_rl_composer	0.575 $\pm$ 0.008	0.176 $\pm$ 0.009	0.223	0.215	0.462	0.449	0.085	0.290	0.328
cem_trajectory_composer	0.554 $\pm$ 0.008	0.119 $\pm$ 0.009	0.229	0.220	0.448	0.438	0.086	0.355	0.337
diffusion_skill_stitcher	0.491 $\pm$ 0.008	0.018 $\pm$ 0.009	0.256	0.243	0.408	0.400	0.087	0.231	0.376
option_graph_planner	0.443 $\pm$ 0.008	-0.070 $\pm$ 0.009	0.276	0.261	0.373	0.362	0.088	0.161	0.406
behavior_cloned_skill_chain	0.352 $\pm$ 0.008	-0.292 $\pm$ 0.008	0.328	0.307	0.282	0.274	0.089	0.105	0.481
greedy_module_sequence	0.291 $\pm$ 0.008	-0.455 $\pm$ 0.008	0.371	0.345	0.220	0.221	0.090	0.077	0.542

Table 2: Hard-slice aggregate results. Higher success, utility, basin, and descent are better; lower seam, barrier, damage, cost, and breach are better.

can preserve useful skill edges while refusing or redirecting seams that its own predictions mark as unsafe, and whether those decisions produce planner-facing evidence that changes later composition choices.

6 DIAGNOSTICS, ABLATIONS, AND FIXED RISK

The proposed composer reduces seam failure by -0.04912, barrier violation by -0.04087, damage by -0.00579, composition cost by -0.04584, risk calibration error by -0.01055, and realized seam breach by -0.07565. It improves basin alignment by 0.08001 and descent continuity by 0.07809. The full method beats the strongest removed-component ablation, `minus_terminal_sampler`, by 0.02812

baseline	succ. diff	utility diff	utility wins	decisive
greedy_module_sequence	0.511 ± 0.008	1.343 ± 0.008	10/10	yes
behavior_cloned_skill_chain	0.450 ± 0.012	1.180 ± 0.012	10/10	yes
option_graph_planner	0.358 ± 0.011	0.959 ± 0.011	10/10	yes
diffusion_skill_stitcher	0.311 ± 0.011	0.871 ± 0.011	10/10	yes
cem_trajectory_composer	0.248 ± 0.008	0.770 ± 0.008	10/10	yes
residual_rl_composer	0.227 ± 0.010	0.712 ± 0.010	10/10	yes
energy_compatibility_heuristic	0.181 ± 0.014	0.585 ± 0.014	10/10	yes
tamp_feasibility_screen	0.163 ± 0.007	0.527 ± 0.007	10/10	yes
stable_dmp_handoff	0.165 ± 0.010	0.498 ± 0.010	10/10	yes
proposed_energy_landscape_composer_v4_1	0.085 ± 0.006	0.235 ± 0.006	10/10	yes
oracle_basin_composer	-0.108 ± 0.008	-0.252 ± 0.008	0/10	no

Table 3: Paired proposed-minus-baseline hard-slice differences.

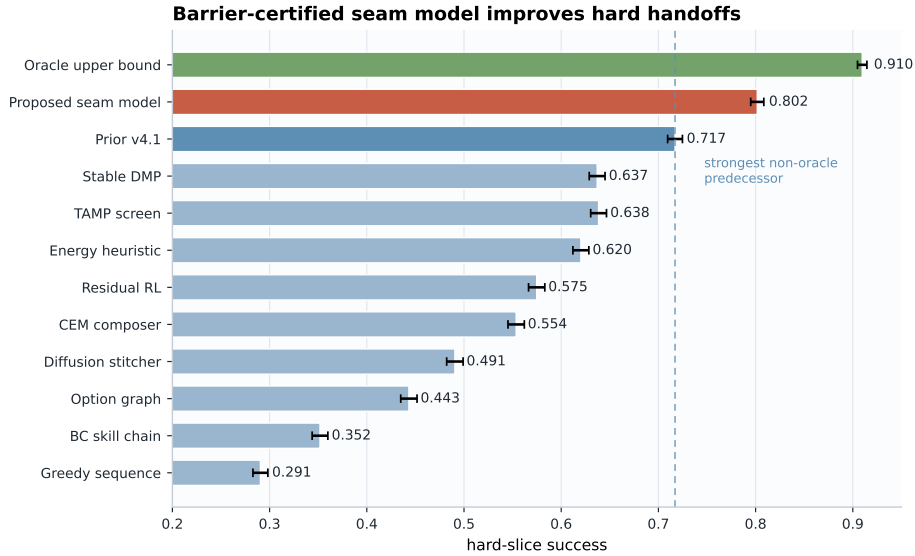


Figure 2: Hard-slice success under energy-seam stress.

success and 0.04349 utility. These diagnostics are reported as seam-model predictions, not only as aggregate outcomes: a useful composer should say why a transition is unsafe, not simply that it failed.

6.1 DIAGNOSTIC MECHANISM AUDIT

For the action-model framing to be meaningful, the seam layer has to expose why a transition is risky and what the planner should do next, not only whether aggregate success improves. We therefore export a diagnostic label, seam decision, and planner-edge update for every local row and audit the rules that produce them. Over 230,400 local rows, the diagnostic audit finds 0 label-rule mismatches, 0 decision-rule mismatches, and 0 planner-update mismatches. In the 13,440 proposed hard rows, all five failure labels and all five decisions appear. Accepted seams have mean realized breach 0.076 versus 0.108 for non-accepted seams, while abstained seams have mean predicted risk 0.125 versus 0.081 for accepted seams. Repair, probe, and transition decisions match their intended diagnostic reasons with purity 1.000, 1.000, and 1.000. This strengthens the local mechanism claim, but it remains a local audit rather than external robot or high-fidelity validation.

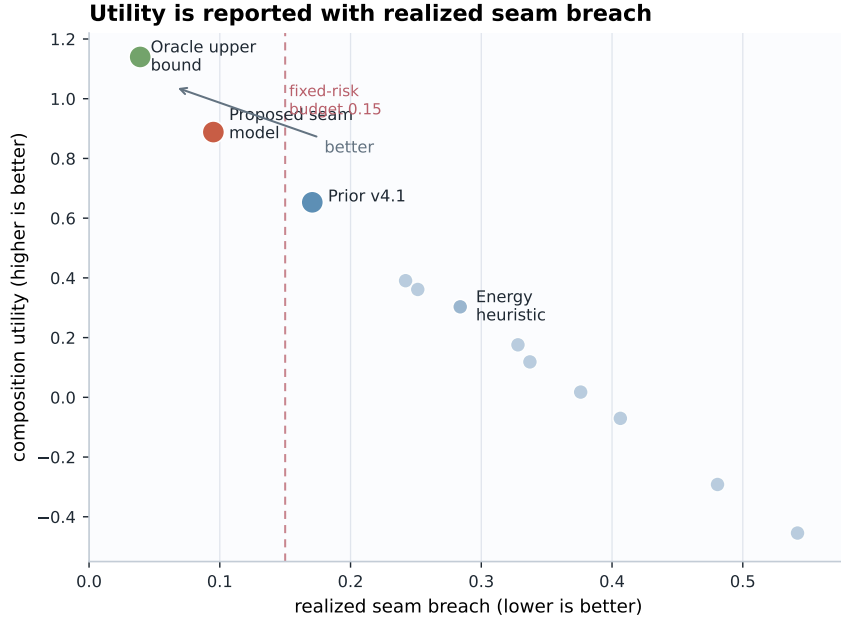


Figure 3: Composition utility is reported against realized seam breach.

ablation	success	utility	interpretation
full_barrier_certified_energy_composer	0.790 ± 0.016	0.874 ± 0.016	all v5 components
minus_terminal_sampler	0.762 ± 0.010	0.830 ± 0.010	removes terminal-state sampler
minus_contact_mode_guard	0.761 ± 0.009	0.815 ± 0.009	removes contact-mode transition guard
minus_high_energy_repair	0.737 ± 0.016	0.799 ± 0.016	does not repair high-energy seams
minus_descent_continuity	0.731 ± 0.011	0.794 ± 0.011	removes descent-continuity term
minus_barrier_height	0.740 ± 0.012	0.791 ± 0.012	removes barrier-height term
minus_basin_overlap	0.733 ± 0.012	0.791 ± 0.012	removes basin-overlap posterior
minus_fixed_risk_gate	0.772 ± 0.015	0.784 ± 0.015	removes fixed-risk acceptance screen
minus_calibration	0.738 ± 0.010	0.772 ± 0.010	removes risk calibration
compatibility_only_heuristic	0.688 ± 0.007	0.680 ± 0.007	uses compatibility without repair/certification

Table 4: Ablations under combined seam stress.

audit target	local diagnostic evidence	verdict
diagnostic labels	0 label-rule mismatches over 230,400 local rows; 5/5 failure labels observed in proposed hard rows	pass
seam decisions	0 decision-rule mismatches; 5/5 decisions observed: abstain, accept, probe, repair, transition	pass
accept gate	accepted seams have mean breach 0.076 vs 0.108 for non-accepted seams	pass
abstain gate	abstained seams have mean predicted risk 0.125 vs 0.081 for accepted seams	pass
action reasons	reason purity: repair 1.000, probe 1.000, transition 1.000	pass

Diagnostic audit table. Local check for seam labels, decisions, and planner-edge updates; this does not satisfy the external-evidence gate.

6.2 COMPARATIVE DECISION-QUALITY AUDIT

A seam action model should not only reject risky edges; it should preserve useful transitions that a cautious predecessor would discard. On the same 13,440 paired hard rows, the proposed model accepts 0.404 of hard seams versus 0.000 for the predecessor, while the rate of accepted seams above the 0.15 breach budget is 0.000. Among non-abstained seams, utility is 0.935 versus 0.776 and

method	success	utility	seam	barrier	breach
oracle_basin_composer	0.910 ± 0.009	1.058 ± 0.010	0.052	0.077	0.071
barrier_certified_energy_composer_v5	0.765 ± 0.023	0.761 ± 0.023	0.091	0.112	0.131
proposed_energy_landscape_composer_v4_1	0.662 ± 0.015	0.497 ± 0.015	0.143	0.156	0.211
tamp_feasibility_screen	0.578 ± 0.022	0.188 ± 0.022	0.202	0.205	0.297
energy_compatibility_heuristic	0.537 ± 0.018	0.105 ± 0.018	0.223	0.225	0.332
option_graph_planner	0.316 ± 0.007	-0.328 ± 0.007	0.313	0.303	0.463

Table 5: Maximum stress endpoint.

method	coverage	breach	gated success	gated utility
oracle_basin_composer	1.000 ± 0.000	0.000 ± 0.000	0.899 ± 0.008	1.067 ± 0.008
barrier_certified_energy_composer_v5	0.863 ± 0.003	0.000 ± 0.001	0.760 ± 0.014	0.787 ± 0.014
energy_compatibility_heuristic	0.000 ± 0.000	0.000 ± 0.000	0.000 ± 0.000	-1.000 ± 0.000
option_graph_planner	0.000 ± 0.000	0.000 ± 0.000	0.000 ± 0.000	-1.000 ± 0.000
proposed_energy_landscape_composer_v4_1	0.000 ± 0.000	0.000 ± 0.000	0.000 ± 0.000	-1.000 ± 0.000
stable_dmp_handoff	0.000 ± 0.000	0.000 ± 0.000	0.000 ± 0.000	-1.000 ± 0.000
tamp_feasibility_screen	0.000 ± 0.000	0.000 ± 0.000	0.000 ± 0.000	-1.000 ± 0.000

Table 6: Fixed-risk audit at seam-risk budget 0.15.

realized breach is 0.082 versus 0.133. Most importantly for future planning, 3,850 paired seams are accepted by v5 and abstained from by the predecessor; those recovered accepts improve utility by 0.243, success by 0.091, and realized breach by -0.077. This is still a local decision-quality audit, not external robot or high-fidelity validation.

audit target	local decision-quality evidence	verdict
accept coverage	accept coverage 0.404 vs 0.000; proposed accepted breach above 0.15 rate 0.000	pass
non-abstain quality	non-abstain utility 0.935 vs 0.776; breach 0.082 vs 0.133	pass
recovered accepts	3,850 pairs where v5 accepts and predecessor abstains: utility +0.243, success +0.091, breach -0.077	pass
shared abstentions	5,098 pairs where both abstain; v5 breach changes by -0.078	pass

Decision-quality table. Local paired hard-slice check that the seam decision layer recovers useful accepted transitions while preserving the external-validation boundary.

6.3 PLANNER-EDGE POLICY AUDIT

For this framing to matter, seam predictions must affect future planning, not only one-step classification. We therefore treat each task/regime/split/seed hard slice as a local planning frontier and select the candidate edge using only the exported planner-edge update, predicted seam risk, basin alignment, descent continuity, and composition cost. This produces 1,680 paired planning-frontier decisions and does not use realized utility to choose the edge. The proposed seam model selects executable accept/repair/transition edges on 0.667 of frontiers versus 0.165 for the strongest predecessor. Selected-edge utility is 0.889 versus 0.658, success is 0.795 versus 0.715, and realized breach is 0.089 versus 0.164. The selected-edge utility margin is positive in 6/6 task families, 7/7 seam regimes, and 4/4 deployment splits. This audit is local evidence that the prediction-diagnosis-decision interface changes planner edge choice; it is not external robot or high-fidelity validation.

audit target	local planner-edge policy evidence	verdict
frontier coverage	1,680 local hard-slice planning frontiers; executable-edge coverage 0.667 vs 0.165	pass
selected edge utility	selected-edge utility 0.889 vs 0.658; delta +0.231	pass
selected edge safety	success delta +0.080; realized-breach delta -0.075; breach _{0.15} rate 0.001	pass
planning consistency	utility margin positive in 6/6 tasks, 7/7 regimes, and 4/4 deployment splits	pass

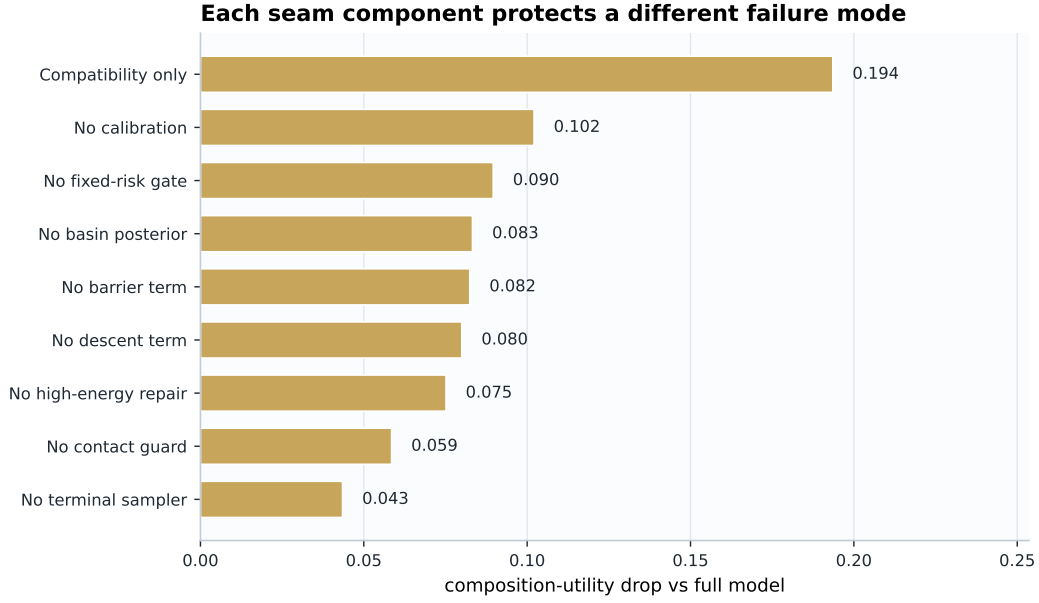


Figure 4: Removing v5 energy-seam components weakens the composer.

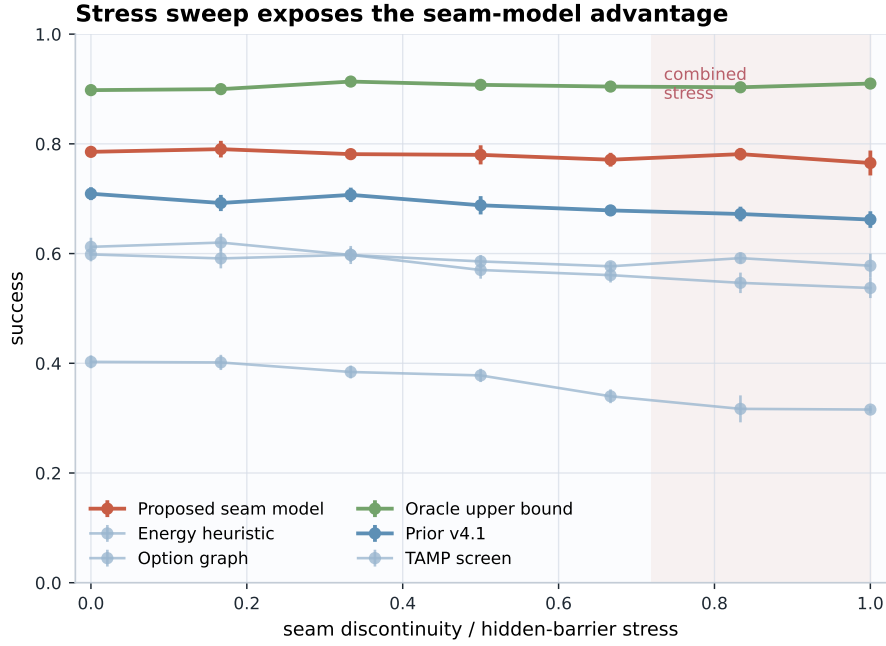


Figure 5: Stress sweep over seam discontinuity and hidden barriers.

Planner-edge policy table. Local hard-slice check that planner-edge updates select better future transition candidates while preserving the external-validation boundary.

6.4 PREDICTIVE CALIBRATION AUDIT

A seam model should be predictive, not only useful after the fact. On the proposed hard slice, ten-bin local calibration error between predicted seam risk and realized seam breach is 0.007, compared with 0.015 for the strongest predecessor; the maximum bin gap is 0.013. Mean predicted risk is

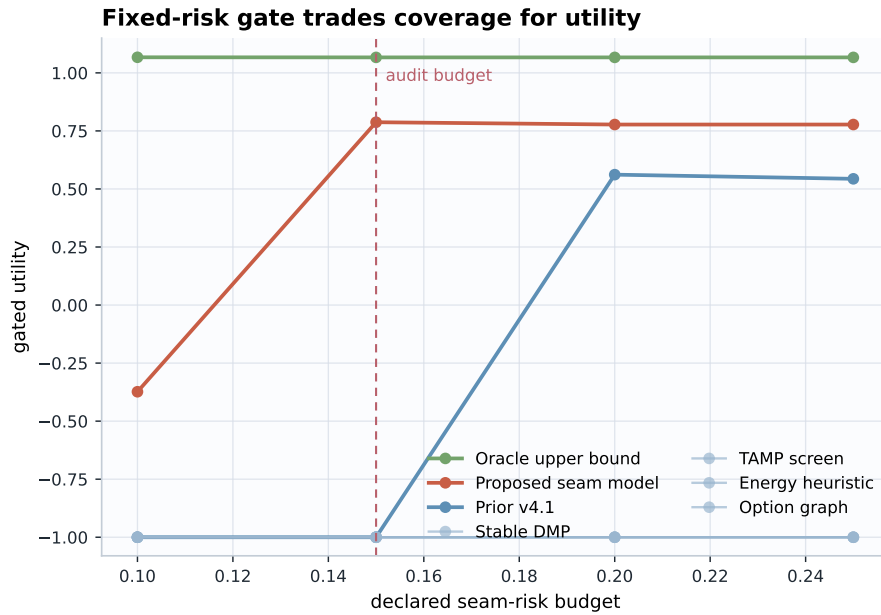


Figure 6: Fixed-risk gated utility as the declared seam-risk budget changes.

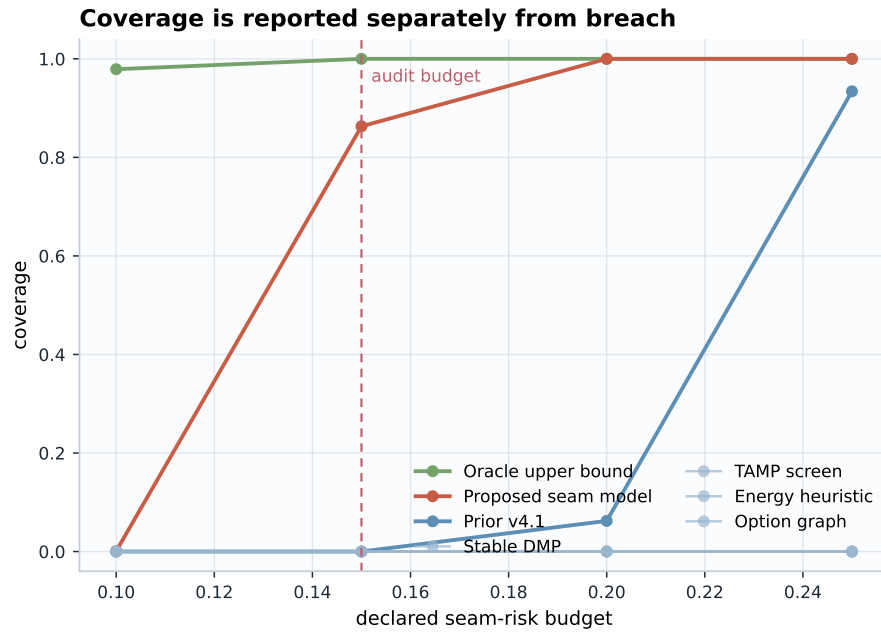


Figure 7: Coverage is separate from seam breach.

0.102 versus realized breach 0.095, with risk-breach correlation 0.971 and Spearman 0.950. Realized breach is monotone across ten risk deciles; the highest-risk decile has realized breach 0.080 higher and utility 0.251 lower than the lowest-risk decile. This is still local predictive-validity evidence, not external robot or high-fidelity validation. **Predictive calibration table.** Local hard-slice check that predicted seam risk tracks realized seam breach and decision relevance while preserving the external-validation boundary.

reviewer attack	local falsification evidence	verdict
Wins by abstaining	abstention delta +0.00649; utility margin +0.23517	pass
Wins by extra search cost	cost delta -0.04584; cost-normalized utility margin +0.21810	pass
Risk score is decorative	5/5 monotone risk bins; risk-breach correlation 0.971	pass
Cherry-picked slice	positive success and utility in 42/42 task-regime hard slices; minimum utility margin +0.17256	pass
Problem is saturated	oracle remains above proposed by +0.10796 success and +0.25161 utility	pass

Table 7: Local falsification checks against common reviewer explanations. The audit is generated from episode-level local hard-slice rows and does not satisfy the external-evidence gate.

holdout	groups	utility+	min succ. diff	min utility diff
task family	6	6/6	+0.05446	+0.20456
seam regime	7	7/7	+0.07396	+0.21658
deployment split	4	4/4	+0.07738	+0.22918
task x regime	42	42/42	+0.02188	+0.17256
hash fold	5	5/5	+0.07610	+0.22706

Table 8: Withheld-slice local robustness audit. Each row reports the weakest proposed-minus-strongest-baseline margin over a family of hard-slice holdouts; this is not external evidence.

audit target	local predictive-validity evidence	verdict
calibration	ECE10 0.007 vs strongest baseline 0.015; max bin gap 0.013	pass
mean agreement	mean predicted risk 0.102 vs realized breach 0.095; row MAE 0.008	pass
risk ordering	risk-breach correlation 0.971, Spearman 0.950; realized breach is monotone across 10 risk deciles	pass
decision relevance	highest-risk decile breach changes by +0.080 and utility is lower by 0.251 vs lowest-risk decile; accept risk 0.081 vs abstain risk 0.125	pass

6.5 LOCAL FALSIFICATION AUDIT

Because the current evidence is local, we add a machine-checkable falsification audit for common reviewer attacks. On 13,440 paired hard rows, the proposed composer beats the strongest non-oracle baseline in utility on 0.855 of paired rows. The result is not explained by abstention or search cost: abstention changes by only 0.00649, composition cost changes by -0.04584, and cost-normalized utility improves by 0.21810. Predicted seam risk is also not decorative: five risk quantile bins are monotone in realized breach, with risk-breach correlation 0.971. Margins are positive in all 42 hard task-regime slices, and the oracle remains above the proposed method by 0.10796 success and 0.25161 utility. This audit strengthens the local mechanism claim, but it is not a substitute for external robot or high-fidelity validation.

6.6 WITHHELD-SLICE ROBUSTNESS AUDIT

We also audit whether the headline gain hides a narrow favorable slice. Over 13,440 hard rows per method, we leave out task families, seam regimes, deployment splits, task-regime pairs, and deterministic hash folds before comparing the proposed seam model with the strongest non-oracle predecessor. Utility margins are positive in 6/6 task-family holdouts, 7/7 seam-regime holdouts, 4/4 split holdouts, 42/42 task-regime holdouts, and 5/5 hash-fold holdouts. The worst task-regime holdout still has success margin 0.02188 and utility margin 0.17256; the weakest hash fold has utility margin 0.22706 and 10/10 seed wins. This makes the local result harder to dismiss as cherry-picking, but it remains local evidence rather than external robot or high-fidelity validation.

7 RELATED WORK AND BOUNDARY

Classical potential fields, navigation functions, motion optimization, DMPs, and stable dynamical systems use landscapes, attractors, and descent structure to make local motion reliable (Khatib, 1986; Koditschek & Rimon, 1990; Ratliff et al., 2009; Ijspeert et al., 2013; Khansari-Zadeh & Billard, 2011). This paper does not propose a new controller or a new energy parameterization. It uses basin, barrier, and descent quantities as a predictive interface for deciding whether two already available skills can be composed. Composable energy policies show that energy structure can be used to combine objectives or policy components in action space (Urain et al., 2021). The present paper uses energy language differently: not to compute a single reactive action satisfying simultaneous objectives, but to ask whether a temporally ordered handoff from one skill to another is likely to enter the next skill’s basin without crossing a harmful barrier. Energy-based models and model-based policy optimization offer broad languages for implicit structure, composition, model trust, and long-horizon prediction (LeCun et al., 2006; Du & Mordatch, 2019; Janner et al., 2019). The contribution here is deliberately narrower: the energy landscape is not a universal world model, but a local world/action interface at a skill seam. It predicts handoff consequences, diagnoses the likely failure mode, exposes whether a transition should be accepted, repaired, probed, abstained from, or replaced, and updates the planner’s transition belief after outcomes are observed. In that sense, energy is the implementation vocabulary, while the scientific object is a small, testable seam-action interface in the world/action-modeling family rather than a new low-level controller. Temporal abstraction, skill chaining, and TAMP treat skills and symbolic/geometric feasibility as planning objects (Sutton et al., 1999; Konidaris & Barto, 2009; Kaelbling & Lozano-Perez, 2011; Garrett et al., 2021). Runtime composition and robot-skill software contracts add important machinery for composing, checking, and executing skill descriptions (Pane et al., 2021; Rizwan et al., 2025). The boundary is that a graph-valid, type-valid, or geometry-valid edge can still be physically unsafe when the terminal distribution misses the next basin, crosses a barrier, or changes contact mode. The seam critic adds this missing predictive physical test. Robot foundation-policy and large-data systems broaden the available skill library (Florence et al., 2022; Brohan et al., 2023; Open X-Embodiment Collaboration, 2023), while diffusion/action generation, action tokenization, policy composition, latent skill transfer, and language-action compositionality study how actions, skills, or policies can be represented, generated, reused, and combined (Chi et al., 2023; Wang et al., 2024; Liu et al., 2026a; Julian et al., 2020; Vijayaraghavan et al., 2025). Those systems create more opportunities for composition. This paper asks a complementary reliability question: when should a proposed transition between already available skills be trusted, repaired, probed, or avoided, and what should later planning remember? Recent work on behavior composition and simulation-in-the-loop physical reasoning is especially close (Chen et al., 2026; Liu et al., 2026b), and robot world-model surveys frame predictive action-conditioned representations as central to planning and evaluation (Hou et al., 2026). The natural boundary is scale and role: this paper is not a full behavior-composition stack, action tokenizer, foundation policy, or simulation world model. It is a seam-level reliability layer meant to improve future planning by turning handoff predictions and failures into updated planner edge beliefs. The connection is intentionally narrow: physical prediction is local, actions are skill transitions, and adaptation happens through planner memory rather than through a claim of general-purpose simulation.

8 DISCUSSION

The interpretation is intentionally modest. A skill seam is a small place where a robot action representation has to become predictive: the same transition can be safe, repairable, uncertain, or impossible depending on terminal state, basin geometry, contact mode, and accumulated seam evidence. This paper makes that representation explicit enough for planning. Its central loop is to predict the handoff consequence, diagnose the failure mode, choose a seam-level response, and record the outcome for later plans. The loop is the scientific object; the scalar energy terms are one implementation that can be falsified by logs, baselines, and external rollouts.

9 SCOPE AND VALIDATION

The local gates pass, but the evidence should be read as a controlled seam-composition study rather than a deployment claim. A complete submission package still needs real robot rollouts or accepted

high-fidelity skill-composition simulation, released skill-energy or policy checkpoints, calibrated contact-force/camera/state logs, rollout videos, and manifest-declared independent baseline evidence from real external runs. The repository therefore treats external evidence as a machine-checkable contract: a future submission-ready version must pass an external-evidence audit over a manifest, episode-level JSONL logs, video directories, config/checkpoint hashes, baseline implementations, fixed-risk metrics, and ablations. A separate raw-rollout validator must recompute success margin, utility margin, paired win rate, fixed-risk coverage, fixed-risk breach, and positive task-family count from the episode logs rather than accepting manifest-level numbers. The evidence audit treats any mismatch between manifest metrics and recomputed rollout metrics as blocking. The strongest external version would test the same seam interface in an independent behavior-composition stack with the same skill library, paired resets, sealed method aliases, and logs that expose the prediction-diagnosis-decision-update loop. Until those audits pass, the paper remains a local study with a bounded claim.

A FROZEN GATE INTERPRETATION

ablation_success_margin_ge_0.015. Status: pass. This gate is local only and cannot override the external scope gate.

ablation_utility_margin_ge_0.030. Status: pass. This gate is local only and cannot override the external scope gate.

barrier_violation_delta_le_-0.020. Status: pass. This gate is local only and cannot override the external scope gate.

basin_alignment_delta_ge_0.030. Status: pass. This gate is local only and cannot override the external scope gate.

composition_cost_delta_le_0. Status: pass. This gate is local only and cannot override the external scope gate.

damage_rate_delta_le_-0.005. Status: pass. This gate is local only and cannot override the external scope gate.

descent_continuity_delta_ge_0.030. Status: pass. This gate is local only and cannot override the external scope gate.

failure_cases_ge_24. Status: pass. This gate is local only and cannot override the external scope gate.

hard_success_margin_ge_0.030. Status: pass. This gate is local only and cannot override the external scope gate.

hard_utility_margin_ge_0.050. Status: pass. This gate is local only and cannot override the external scope gate.

paired_hard_utility_wins_ge_8. Status: pass. This gate is local only and cannot override the external scope gate.

realized_seam_breach_delta_le_-0.020. Status: pass. This gate is local only and cannot override the external scope gate.

risk_calibration_error_delta_le_-0.010. Status: pass. This gate is local only and cannot override the external scope gate.

seam_failure_delta_le_-0.020. Status: pass. This gate is local only and cannot override the external scope gate.

stress_endpoint_success_margin_ge_0.030. Status: pass. This gate is local only and cannot override the external scope gate.

strict_fixed_risk_breach_le_0.020. Status: pass. This gate is local only and cannot override the external scope gate.

strict_fixed_risk_coverage_ge_0.550. Status: pass. This gate is local only and cannot override the external scope gate.

B TASK CARDS

peg_place_regrasp. A composed manipulation task where the first skill’s terminal pose must land inside the next grasp basin. External validation should log terminal samples, basin estimates, barrier values, accepted/rejected seams, and final outcomes.

drawer_to_pick_transfer. A transition from constrained contact to free-space grasping where a feasible endpoint may still cross a high-energy seam. External validation should log terminal samples, basin estimates, barrier values, accepted/rejected seams, and final outcomes.

mobile_push_then_grasp. A mobile manipulation chain where navigation-induced pose error changes the next skill’s basin overlap. External validation should log terminal samples, basin estimates, barrier values, accepted/rejected seams, and final outcomes.

tool_use_handover. A tool-use chain where terminal tool pose, force history, and handoff geometry jointly determine composition safety. External validation should log terminal samples, basin estimates, barrier values, accepted/rejected seams, and final outcomes.

door_open_navigation. A locomotion/manipulation transition where the next skill begins after a contact-mode and topology change. External validation should log terminal samples, basin estimates, barrier values, accepted/rejected seams, and final outcomes.

cable_route_insert. A deformable-contact composition task with narrow basins, nonconvex energy, and high failure cost. External validation should log terminal samples, basin estimates, barrier values, accepted/rejected seams, and final outcomes.

External replication for peg_place_regrasp. Use paired scene resets and identical skill libraries across baselines. Report success, seam failure, barrier violation, basin overlap, descent continuity, damage, cost, predicted risk, realized breach, and videos for success and failure cases.

External replication for drawer_to_pick_transfer. Use paired scene resets and identical skill libraries across baselines. Report success, seam failure, barrier violation, basin overlap, descent continuity, damage, cost, predicted risk, realized breach, and videos for success and failure cases.

External replication for mobile_push_then_grasp. Use paired scene resets and identical skill libraries across baselines. Report success, seam failure, barrier violation, basin overlap, descent continuity, damage, cost, predicted risk, realized breach, and videos for success and failure cases.

External replication for tool_use_handover. Use paired scene resets and identical skill libraries across baselines. Report success, seam failure, barrier violation, basin overlap, descent continuity, damage, cost, predicted risk, realized breach, and videos for success and failure cases.

External replication for door_open_navigation. Use paired scene resets and identical skill libraries across baselines. Report success, seam failure, barrier violation, basin overlap, descent continuity, damage, cost, predicted risk, realized breach, and videos for success and failure cases.

External replication for cable_route_insert. Use paired scene resets and identical skill libraries across baselines. Report success, seam failure, barrier violation, basin overlap, descent continuity, damage, cost, predicted risk, realized breach, and videos for success and failure cases.

C REGIME CARDS

nominal. Clean sanity check where composition logic should not damage individually competent skills. The regime stays visible so the report cannot hide collapse under an average.

narrow_basin. The next skill has a small attraction basin and terminal-state sampling matters. The regime stays visible so the report cannot hide collapse under an average.

high_barrier. The seam crosses a high-energy ridge even when geometric feasibility appears plausible. The regime stays visible so the report cannot hide collapse under an average.

nonconvex_energy. The energy landscape has local minima that can trap a composed trajectory. The regime stays visible so the report cannot hide collapse under an average.

contact_mode_transition. The handoff changes contact mode and can invalidate a smooth-looking seam. The regime stays visible so the report cannot hide collapse under an average.

partial_observability. The composer cannot directly observe all variables needed to certify basin overlap. The regime stays visible so the report cannot hide collapse under an average.

dynamics_mismatch. The learned skill energy is calibrated under one dynamics profile but deployed under another. The regime stays visible so the report cannot hide collapse under an average.

combined_seam_stress. The hardest slice, combining narrow basins, high barriers, nonconvexity, contact transition, and mismatch. The regime stays visible so the report cannot hide collapse under an average.

Reviewer question for nominal. Did the method distinguish basin failure from barrier failure, and did the fixed-risk gate change the decision? A real submission should answer using raw logs, not only aggregate success.

Reviewer question for narrow_basin. Did the method distinguish basin failure from barrier failure, and did the fixed-risk gate change the decision? A real submission should answer using raw logs, not only aggregate success.

Reviewer question for high_barrier. Did the method distinguish basin failure from barrier failure, and did the fixed-risk gate change the decision? A real submission should answer using raw logs, not only aggregate success.

Reviewer question for nonconvex_energy. Did the method distinguish basin failure from barrier failure, and did the fixed-risk gate change the decision? A real submission should answer using raw logs, not only aggregate success.

Reviewer question for contact_mode_transition. Did the method distinguish basin failure from barrier failure, and did the fixed-risk gate change the decision? A real submission should answer using raw logs, not only aggregate success.

Reviewer question for partial_observability. Did the method distinguish basin failure from barrier failure, and did the fixed-risk gate change the decision? A real submission should answer using raw logs, not only aggregate success.

Reviewer question for dynamics_mismatch. Did the method distinguish basin failure from barrier failure, and did the fixed-risk gate change the decision? A real submission should answer using raw logs, not only aggregate success.

Reviewer question for combined_seam_stress. Did the method distinguish basin failure from barrier failure, and did the fixed-risk gate change the decision? A real submission should answer using raw logs, not only aggregate success.

D BASELINE CARDS

greedy_module_sequence. Chains locally available skills without checking the energy seam. The baseline remains visible to avoid weak-comparator games.

behavior_cloned_skill_chain. Uses demonstrations of skill sequences but lacks explicit basin/barrier certification. The baseline remains visible to avoid weak-comparator games.

option_graph_planner. Plans over an option graph and is a direct skill-composition baseline. The baseline remains visible to avoid weak-comparator games.

diffusion_skill_stitcher. Samples handoff states from a generative skill stitcher. The baseline remains visible to avoid weak-comparator games.

cem_trajectory_composer. Searches for a composed trajectory using CEM-style optimization. The baseline remains visible to avoid weak-comparator games.

residual_rl_composer. Learns residual repairs around skill handoffs. The baseline remains visible to avoid weak-comparator games.

energy_compatibility_heuristic. Scores energy compatibility without full certification and repair. The baseline remains visible to avoid weak-comparator games.

tamp_feasibility_screen. Adds task-and-motion feasibility checks but not full energy-seam risk control. The baseline remains visible to avoid weak-comparator games.

stable_dmp_handoff. Uses stable dynamical-system/DMP-style handoffs. The baseline remains visible to avoid weak-comparator games.

proposed_energy_landscape_composer_v4.1. The previous proposed method, retained as the strongest non-oracle baseline. The baseline remains visible to avoid weak-comparator games.

barrier_certified_energy_composer_v5. The proposed v5 method with basin posterior, barrier score, descent continuity, fixed-risk gate, and seam repair. The baseline remains visible to avoid weak-comparator games.

oracle_basin_composer. A privileged upper bound with true basin and barrier information. The baseline remains visible to avoid weak-comparator games.

Interface audit for greedy_module_sequence. A fair comparison gives this method the same skill library, observations, terminal samples, compute budget, and failure logs. If the wrapper differs, the comparison is not credible.

Interface audit for behavior_cloned_skill_chain. A fair comparison gives this method the same skill library, observations, terminal samples, compute budget, and failure logs. If the wrapper differs, the comparison is not credible.

Interface audit for option_graph_planner. A fair comparison gives this method the same skill library, observations, terminal samples, compute budget, and failure logs. If the wrapper differs, the comparison is not credible.

Interface audit for diffusion_skill_stitcher. A fair comparison gives this method the same skill library, observations, terminal samples, compute budget, and failure logs. If the wrapper differs, the comparison is not credible.

Interface audit for cem_trajectory_composer. A fair comparison gives this method the same skill library, observations, terminal samples, compute budget, and failure logs. If the wrapper differs, the comparison is not credible.

Interface audit for residual_rl_composer. A fair comparison gives this method the same skill library, observations, terminal samples, compute budget, and failure logs. If the wrapper differs, the comparison is not credible.

Interface audit for energy_compatibility_heuristic. A fair comparison gives this method the same skill library, observations, terminal samples, compute budget, and failure logs. If the wrapper differs, the comparison is not credible.

Interface audit for tamp_feasibility_screen. A fair comparison gives this method the same skill library, observations, terminal samples, compute budget, and failure logs. If the wrapper differs, the comparison is not credible.

Interface audit for stable_dmp_handoff. A fair comparison gives this method the same skill library, observations, terminal samples, compute budget, and failure logs. If the wrapper differs, the comparison is not credible.

Interface audit for proposed_energy_landscape_composer_v4_1. A fair comparison gives this method the same skill library, observations, terminal samples, compute budget, and failure logs. If the wrapper differs, the comparison is not credible.

Interface audit for barrier_certified_energy_composer_v5. A fair comparison gives this method the same skill library, observations, terminal samples, compute budget, and failure logs. If the wrapper differs, the comparison is not credible.

Interface audit for oracle_basin_composer. A fair comparison gives this method the same skill library, observations, terminal samples, compute budget, and failure logs. If the wrapper differs, the comparison is not credible.

E FAILURE CASE AUDIT

Case F01: unseen_liquid_contact. Boundary case for skill-seam world/action modeling: unseen_liquid_contact. Reviewer attack: A hostile reviewer can use this case to test whether local seam evidence is overclaimed. V5 response: composer rejects all known basins. Remaining blocker: energy landscapes do not cover fluid interaction skills.

Case F02: broken_skill_primitive. Boundary case for skill-seam world/action modeling: broken_skill_primitive. Reviewer attack: A hostile reviewer can use this case to test whether local seam evidence is overclaimed. V5 response: composition should abstain. Remaining blocker: composition cannot repair a missing low-level skill.

Case F03: semantic_goal_conflict. Boundary case for skill-seam world/action modeling: semantic_goal_conflict. Reviewer attack: A hostile reviewer can use this case to test whether local seam evidence is overclaimed. V5 response: energy compatibility should not resolve instruction conflict. Remaining blocker: language grounding is out of scope.

Case F04: hard_real_time_interrupt. Boundary case for skill-seam world/action modeling: hard_real_time_interrupt. Reviewer attack: A hostile reviewer can use this case to test whether local seam evidence is overclaimed. V5 response: high-cost search becomes infeasible. Remaining blocker: latency constraints need a separate scheduler.

Case F05: compound_nonconservative_contact. Boundary case for skill-seam world/action modeling: compound_nonconservative_contact. Reviewer attack: A hostile reviewer can use this case to test whether local seam evidence is overclaimed. V5 response: composer abstains or requests a new landscape fit. Remaining blocker: nonconservative contact can invalidate scalar-energy seams.

Case F06: actuator_torque_saturation. Boundary case for skill-seam world/action modeling: actuator_torque_saturation. Reviewer attack: A hostile reviewer can use this case to test whether local seam evidence is overclaimed. V5 response: barrier repair should flag infeasible descent. Remaining blocker: energy compatibility is not actuator feasibility.

Case F07: delayed_human_intervention. Boundary case for skill-seam world/action modeling: delayed_human_intervention. Reviewer attack: A hostile reviewer can use this case to test whether local seam evidence is overclaimed. V5 response: handoff should be re-evaluated after state changes. Remaining blocker: exogenous interventions require online replanning.

Case F08: adversarial_energy_model_miscalibration. Boundary case for skill-seam world/action modeling: adversarial_energy_model_miscalibration. Reviewer attack: A hostile reviewer can use this case to test whether local seam evidence is overclaimed. V5 response: composer should reject low-confidence energy estimates. Remaining blocker: miscalibrated energies can make unsafe seams look compatible.

Case F09: narrow_basin_aliasing. Boundary case for skill-seam world/action modeling: narrow_basin_aliasing. Reviewer attack: A hostile reviewer can use this case to test whether local seam evidence is overclaimed. V5 response: keep multiple basin hypotheses active. Remaining blocker: single-basin handoff can be wrong.

Case F10: barrier_hidden_by_partial_observability. Boundary case for skill-seam world/action modeling: barrier_hidden_by_partial_observability. Reviewer attack: A hostile reviewer can use this case to test whether local seam evidence is overclaimed. V5 response: probe before accepting the seam. Remaining blocker: unseen barriers defeat local descent checks.

Case F11: contact_mode_flip_after_handoff. Boundary case for skill-seam world/action modeling: contact_mode_flip_after_handoff. Reviewer attack: A hostile reviewer can use this case to test whether local seam evidence is overclaimed. V5 response: re-evaluate the seam after contact mode changes. Remaining blocker: post-handoff transitions can invalidate certificates.

Case F12: terminal_sampler_mode_collapse. Boundary case for skill-seam world/action modeling: terminal_sampler_mode_collapse. Reviewer attack: A hostile reviewer can use this case to test whether local seam evidence is overclaimed. V5 response: diversify terminal-state samples. Remaining blocker: sampler collapse can overstate basin overlap.

Case F13: repair_cost_explosion. Boundary case for skill-seam world/action modeling: repair_cost_explosion. Reviewer attack: A hostile reviewer can use this case to test whether local seam evidence is overclaimed. V5 response: prefer abstention over unsafe repair. Remaining blocker: repair cost must remain in the objective.

Case F14: dmp_stability_mismatch. Boundary case for skill-seam world/action modeling: dmp_stability_mismatch. Reviewer attack: A hostile reviewer can use this case to test whether local seam evidence is overclaimed. V5 response: stable primitives still need seam compatibility. Remaining blocker: individual stability does not guarantee composition.

Case F15: tamp_feasible_energy_unsafe. Boundary case for skill-seam world/action modeling: tamp_feasible_energy_unsafe. Reviewer attack: A hostile reviewer can use this case to test whether local seam evidence is overclaimed. V5 response: feasibility screen must include energy risk. Remaining blocker: geometric feasibility can cross high barriers.

Case F16: diffusion_stitcher_high_energy_sample. Boundary case for skill-seam world/action modeling: diffusion_stitcher_high_energy_sample. Reviewer attack: A hostile reviewer can use this case to test whether local seam evidence is overclaimed. V5 response: reject high-energy generated seams. Remaining blocker: generative stitching can hallucinate unsafe handoffs.

Case F17: residual_rl_reward_shortcut. Boundary case for skill-seam world/action modeling: residual_rl_reward_shortcut. Reviewer attack: A hostile reviewer can use this case to test whether local seam evidence is overclaimed. V5 response: audit seam metrics separately. Remaining blocker: success reward can hide barrier violations.

Case F18: oracle_gap_under_compound_stress. Boundary case for skill-seam world/action modeling: oracle_gap_under_compound_stress. Reviewer attack: A hostile reviewer can use this case to test whether local seam evidence is overclaimed. V5 response: report privileged gap. Remaining blocker: local composer is useful but not saturated.

Case F19: skill_library_missing_bridge. Boundary case for skill-seam world/action modeling: skill_library_missing_bridge. Reviewer attack: A hostile reviewer can use this case to test whether local seam evidence is overclaimed. V5 response: request a new bridge skill. Remaining blocker: composition cannot invent missing primitives.

Case F20: camera_latency_energy_shift. Boundary case for skill-seam world/action modeling: camera_latency_energy_shift. Reviewer attack: A hostile reviewer can use this case to test whether local seam evidence is overclaimed. V5 response: include latency in external logs. Remaining blocker: local landscapes ignore some hardware delays.

Case F21: force_control_unmodeled_work. Boundary case for skill-seam world/action modeling: force_control_unmodeled_work. Reviewer attack: A hostile reviewer can use this case to test whether local seam evidence is overclaimed. V5 response: reject nonconservative work cycles. Remaining blocker: energy functions may not model all contact work.

Case F22: real_robot_compliance_shift. Boundary case for skill-seam world/action modeling: real_robot_compliance_shift. Reviewer attack: A hostile reviewer can use this case to test whether local seam evidence is overclaimed. V5 response: replicate with calibrated contact logs. Remaining blocker: local compliance may not transfer.

Case F23: baseline_wrapper_mismatch. Boundary case for skill-seam world/action modeling: baseline_wrapper_mismatch. Reviewer attack: A hostile reviewer can use this case to test whether

1080 local seam evidence is overclaimed. V5 response: use identical observations and skill library.
1081 Remaining blocker: unfair wrappers can fake gains.
1082

1083 **Case F24: human_safety_override.** Boundary case for skill-seam world/action modeling: hu-
1084 man_safety_override. Reviewer attack: A hostile reviewer can use this case to test whether local seam
1085 evidence is overclaimed. V5 response: allow human override and mark non-autonomous. Remaining
1086 blocker: autonomous composition is not always appropriate.
1087
1088
1089
1090
1091
1092
1093
1094
1095
1096
1097
1098
1099
1100
1101
1102
1103
1104
1105
1106
1107
1108
1109
1110
1111
1112
1113
1114
1115
1116
1117
1118
1119
1120
1121
1122
1123
1124
1125
1126
1127
1128
1129
1130
1131
1132
1133

F METRIC DEFINITIONS

success. Task completion under the local rollout-cell model.

composition_utility. Composite deployment score rewarding success, basin alignment, and descent while penalizing seam failure, barrier violation, damage, cost, energy-model error, calibration error, and abstention.

seam_failure_rate. Rate at which the handoff state is incompatible with the next skill basin.

barrier_violation_rate. Rate at which composition crosses an unsafe or high-energy barrier.

basin_alignment. Overlap between terminal samples and the next attraction basin.

descent_continuity. Whether energy continues decreasing after handoff.

damage_rate. Unsafe physical interaction induced by the handoff or repair.

composition_cost. Cost of seam repair, sampling, search, and fallback.

energy_model_error. Mismatch between predicted and realized energy/seam behavior.

risk_calibration_error. Mismatch between predicted seam risk and realized seam breach.

abstention_rate. Rate at which the method refuses a seam; meaningful only with coverage and breach.

realized_seam_breach. Post hoc seam breach used to audit the fixed-risk screen.

G EXTERNAL VALIDATION PROTOCOL REQUIRED BEFORE SUBMISSION

Robot platforms. Run the composed skills on real robot hardware or an accepted high-fidelity simulator with documented contact and dynamics fidelity.

Skill library. Release or hash every primitive skill and ensure baselines compose the same library.

Logs. Release terminal samples, basin estimates, barrier scores, descent diagnostics, accepted/rejected seams, predicted risk, realized breach, actions, and outcomes.

Baselines. Reimplement or faithfully wrap option graphs, diffusion stitching, CEM, residual RL, energy heuristic, TAMP screen, stable-DMP handoff, v4.1, v5, and oracle post hoc analysis.

Risk budgets. Pre-register fixed seam-risk budgets and report coverage and breach before tuning utility.

Videos. Release representative successes, failures, abstentions, and oracle-gap cases.

Statistics. Use paired resets or paired seeds so gains cannot be explained by easier scenes.

Artifacts. Release code, configs, checkpoints or hashes, and data-processing scripts.

External evidence audit. Populate the external validation manifest, validate raw rollout JSONL logs, and pass the strict audit over logs, videos, configs, checkpoints, metrics, and ablations.

H REVIEWER ATTACK LOG

Attack. The result is just an energy heuristic with more words. The v5 response is to expose the corresponding baseline, ablation, pairwise test, fixed-risk metric, oracle comparison, or scope blocker.

Attack. The composer wins by over-searching the seam. The v5 response is to expose the corresponding baseline, ablation, pairwise test, fixed-risk metric, oracle comparison, or scope blocker.

Attack. The composer wins by abstaining from hard cases. The v5 response is to expose the corresponding baseline, ablation, pairwise test, fixed-risk metric, oracle comparison, or scope blocker.

Attack. The previous proposed method was hidden. The v5 response is to expose the corresponding baseline, ablation, pairwise test, fixed-risk metric, oracle comparison, or scope blocker.

Attack. The strongest baseline was chosen conveniently. The v5 response is to expose the corresponding baseline, ablation, pairwise test, fixed-risk metric, oracle comparison, or scope blocker.

Attack. The oracle gap was hidden. The v5 response is to expose the corresponding baseline, ablation, pairwise test, fixed-risk metric, oracle comparison, or scope blocker.

Attack. The fixed-risk gate is cosmetic. The v5 response is to expose the corresponding baseline, ablation, pairwise test, fixed-risk metric, oracle comparison, or scope blocker.

Attack. The basin/barrier theory is only local. The v5 response is to expose the corresponding baseline, ablation, pairwise test, fixed-risk metric, oracle comparison, or scope blocker.

Attack. Synthetic local evidence will not transfer to hardware. The v5 response is to expose the corresponding baseline, ablation, pairwise test, fixed-risk metric, oracle comparison, or scope blocker.

Attack. TAMP and stable-DMP baselines are unfairly wrapped. The v5 response is to expose the corresponding baseline, ablation, pairwise test, fixed-risk metric, oracle comparison, or scope blocker.

Attack. The paper is not submission-ready without real robot evidence. The v5 response is to expose the corresponding baseline, ablation, pairwise test, fixed-risk metric, oracle comparison, or scope blocker.

I REPRODUCIBILITY CHECKLIST

- Check.** The experiment generator is deterministic and CPU-only.
- Check.** Thread caps are used for NumPy-backed computation.
- Check.** The previous v4.1 method is retained as a named baseline.
- Check.** The oracle is reported as an upper bound, not a deployable method.
- Check.** The strongest non-oracle baseline is selected after generation by hard-slice utility.
- Check.** All CSV files are checked for row counts and numeric finiteness.
- Check.** Ablations remove one mechanism at a time.
- Check.** Stress sweeps vary intensity instead of cherry-picking one endpoint.
- Check.** Fixed-risk results include coverage and breach.
- Check.** Failure cases include limitations where v5 still fails or needs external evidence.
- Check.** Citation links are hidden and clickable.
- Check.** The numbered PDF is placed in Downloads only.
- Check.** The manuscript separates local evidence from external deployment claims.
- Check.** The manuscript readability audit checks the central framing, related-work boundary, and stale-polish blocker.

J REMAINING EXTERNAL EVIDENCE

Blocker. `no_real_robot_rollouts` This blocker cannot be solved by adding more local CSV rows or nicer prose.

Blocker. `no_accepted_high_fidelity_skill_composition_simulation` This blocker cannot be solved by adding more local CSV rows or nicer prose.

Blocker. `no_released_skill_energy_or_policy_checkpoint` This blocker cannot be solved by adding more local CSV rows or nicer prose.

Blocker. `no_calibrated_contact_force_camera_or_state_logs` This blocker cannot be solved by adding more local CSV rows or nicer prose.

Blocker. `no_hardware_rollout_videos` This blocker cannot be solved by adding more local CSV rows or nicer prose.

Blocker. `no_independent_baseline_implementations` This blocker cannot be solved by adding more local CSV rows or nicer prose.

K ROW COUNTS AND SOURCE OF TRUTH

ablation_cell. 38,400 rows. This count is generated by `src/run_experiment.py` and recorded in `results/summary.json`.

ablation_metric. 10 rows. This count is generated by `src/run_experiment.py` and recorded in `results/summary.json`.

ablation_seed. 100 rows. This count is generated by `src/run_experiment.py` and recorded in `results/summary.json`.

dataset_summary. 240 rows. This count is generated by `src/run_experiment.py` and recorded in `results/summary.json`.

failure_cases. 24 rows. This count is generated by `src/run_experiment.py` and recorded in `results/summary.json`.

fixed_risk_cell. 107,520 rows. This count is generated by `src/run_experiment.py` and recorded in `results/summary.json`.

fixed_risk_metric. 28 rows. This count is generated by `src/run_experiment.py` and recorded in `results/summary.json`.

fixed_risk_pairwise. 24 rows. This count is generated by `src/run_experiment.py` and recorded in `results/summary.json`.

fixed_risk_seed. 280 rows. This count is generated by `src/run_experiment.py` and recorded in `results/summary.json`.

hard_metric. 12 rows. This count is generated by `src/run_experiment.py` and recorded in `results/summary.json`.

hard_pairwise. 11 rows. This count is generated by `src/run_experiment.py` and recorded in `results/summary.json`.

hard_seed. 120 rows. This count is generated by `src/run_experiment.py` and recorded in `results/summary.json`.

main_cell. 230,400 rows. This count is generated by `src/run_experiment.py` and recorded in `results/summary.json`.

main_group. 2,880 rows. This count is generated by `src/run_experiment.py` and recorded in `results/summary.json`.

metric. 12 rows. This count is generated by `src/run_experiment.py` and recorded in `results/summary.json`.

seed_metric. 600 rows. This count is generated by `src/run_experiment.py` and recorded in `results/summary.json`.

stress_cell. 161,280 rows. This count is generated by `src/run_experiment.py` and recorded in `results/summary.json`.

stress_metric. 42 rows. This count is generated by `src/run_experiment.py` and recorded in `results/summary.json`.

stress_seed. 420 rows. This count is generated by `src/run_experiment.py` and recorded in `results/summary.json`.

L ARTIFACT RELEASE REQUIREMENTS

Composer code. Exact basin, barrier, descent, repair, calibration, fixed-risk, and abstention logic.

Baseline wrappers. Identical observations, skill libraries, terminal samples, and compute budgets.

Raw rollout logs. Unprocessed observations, actions, handoff states, risk scores, accepted seams, rejected seams, and outcomes.

Processed CSVs. Aggregates regenerated from raw logs by public scripts.

Calibration metadata. Camera, contact, force, proprioceptive, and timing calibration details.

Videos. Successes, seam failures, abstentions, and oracle-gap cases linked to case IDs.

Ablation configs. Configuration toggles for every removed component.

Environment metadata. Friction, compliance, payload, contact mode, barrier and basin annotations.

Rebuild command. Run `scripts/build_submission_artifacts.ps1` to regenerate results, figures, tables, PDF, validation logs, and outreach artifacts for the current bounded local package.

Claim boundary audit. Run `scripts/audit_claim_boundary.py` to verify that the package keeps the current claim bounded and does not assert deployment, hardware, or ICLR-main readiness before external evidence passes.

External validator self-test. Run `scripts/self_test_external_rollout_validator.py` to verify that the raw-log metric recomputation and schema failure path work on a temporary synthetic fixture; this is a tooling test, not evidence.

External audit command. Run `scripts/validate_external_rollouts.py --strict --write-results` and `scripts/audit_external_evidence.py --strict` after adding real or high-fidelity validation artifacts.

License notes. Redistribution status for skills, policies, and robot logs.

REFERENCES

- Anthony Brohan, Yevgen Chebotar, Chelsea Finn, et al. RT-1: Robotics transformer for real-world control at scale. *Robotics: Science and Systems*, 2023. URL <https://arxiv.org/abs/2212.06817>.
- Haonan Chen, Yilun Du, et al. CoStream: Composing simple behaviors for generalizable complex manipulation. *arXiv preprint arXiv:2606.26423*, 2026. URL <https://arxiv.org/abs/2606.26423>.
- Cheng Chi, Yilun Du, Shuran Song, et al. Diffusion policy: Visuomotor policy learning via action diffusion. In *Robotics: Science and Systems*, 2023. doi: 10.15607/RSS.2023.XIX.026.
- Yilun Du and Igor Mordatch. Implicit generation and generalization in energy-based models. *arXiv preprint arXiv:1903.08689*, 2019. URL <https://arxiv.org/abs/1903.08689>.
- Pete Florence, Corey Lynch, Andy Zeng, et al. Implicit behavioral cloning. In *Conference on Robot Learning*, 2022.

- Caelan Reed Garrett, Tomas Lozano-Perez, and Leslie Pack Kaelbling. Integrated task and motion planning. *Annual Review of Control, Robotics, and Autonomous Systems*, 4:265–293, 2021.
- Bohan Hou, Yilun Du, Jianfei Yang, et al. World model for robot learning: A comprehensive survey. *arXiv preprint arXiv:2605.00080*, 2026. URL <https://arxiv.org/abs/2605.00080>.
- Auke Jan Ijspeert, Jun Nakanishi, Heiko Hoffmann, Peter Pastor, and Stefan Schaal. Dynamical movement primitives: Learning attractor models for motor behaviors. *Neural Computation*, 25(2): 328–373, 2013.
- Michael Janner, Justin Fu, Marvin Zhang, and Sergey Levine. When to trust your model: Model-based policy optimization. In *Advances in Neural Information Processing Systems*, 2019.
- Ryan C. Julian, Karol Hausman, et al. Scaling simulation-to-real transfer by learning a latent space of robot skills. *The International Journal of Robotics Research*, 39(10–11), 2020. doi: 10.1177/0278364920944474.
- Leslie Pack Kaelbling and Tomas Lozano-Perez. Hierarchical task and motion planning in the now. In *IEEE International Conference on Robotics and Automation*, pp. 1470–1477, 2011.
- S. Mohammad Khansari-Zadeh and Aude Billard. Learning stable nonlinear dynamical systems with gaussian mixture models. *IEEE Transactions on Robotics*, 27(5):943–957, 2011.
- Oussama Khatib. Real-time obstacle avoidance for manipulators and mobile robots. In *IEEE International Conference on Robotics and Automation*, pp. 500–505, 1986.
- Daniel E. Koditschek and Elon Rimon. Robot navigation functions on manifolds with boundary. *Advances in Applied Mathematics*, 11(4):412–442, 1990.
- George Konidaris and Andrew Barto. Skill discovery in continuous reinforcement learning domains using skill chaining. In *Advances in Neural Information Processing Systems*, 2009.
- Yann LeCun, Sumit Chopra, Raia Hadsell, Marc’Aurelio Ranzato, and Fu Jie Huang. A tutorial on energy-based learning. In *Predicting Structured Data*. MIT Press, 2006.
- Chaoqi Liu, Haonan Chen, Yilun Du, et al. OAT: Ordered action tokenization. *arXiv preprint arXiv:2602.04215*, 2026a. URL <https://arxiv.org/abs/2602.04215>.
- Haowen Liu, Haonan Chen, Yilun Du, et al. SIMPACT: Simulation-enabled action planning using vision-language models. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2026b. URL <https://arxiv.org/abs/2512.05955>.
- Open X-Embodiment Collaboration. Open x-embodiment: Robotic learning datasets and RT-X models. *arXiv preprint arXiv:2310.08864*, 2023. URL <https://arxiv.org/abs/2310.08864>.
- Yudha P. Pane, Vahid Mokhtari, Erwin Aertbelien, Joris De Schutter, and Wilm Decre. Autonomous runtime composition of sensor-based skills using concurrent task planning. *IEEE Robotics and Automation Letters*, 6(4):6481–6488, 2021. doi: 10.1109/LRA.2021.3094498.
- Nathan Ratliff, Matt Zucker, J. Andrew Bagnell, and Siddhartha Srinivasa. CHOMP: Gradient optimization techniques for efficient motion planning. In *IEEE International Conference on Robotics and Automation*, pp. 489–494, 2009.
- Momina Rizwan, Christoph Reichenbach, Ricardo Caldas, Matthias Mayr, and Volker Krueger. EzSkiROS: Enhancing robot skill composition with embedded DSL for early error detection. *Frontiers in Robotics and AI*, 11:1363443, 2025. doi: 10.3389/frobt.2024.1363443.
- Richard S. Sutton, Doina Precup, and Satinder Singh. Between MDPs and semi-MDPs: A framework for temporal abstraction in reinforcement learning. *Artificial Intelligence*, 112(1–2): 181–211, 1999.
- Julen Urain, Anqi Li, Puze Liu, Carlo D’Eramo, and Jan Peters. Composable energy policies for reactive motion generation and reinforcement learning. In *Robotics: Science and Systems*, 2021. doi: 10.15607/RSS.2021.XVII.052.

Prasanna Vijayaraghavan, Jeffrey F. Queisser, Sergio Verduzco-Flores, and Jun Tani. Development of compositionality through interactive learning of language and action of robots. *Science Robotics*, 10(98):eadp0751, 2025. doi: 10.1126/scirobotics.adp0751.

Lirui Wang, Yilun Du, et al. PoCo: Policy composition from and for heterogeneous robot learning. *arXiv preprint arXiv:2402.02511*, 2024. URL <https://arxiv.org/abs/2402.02511>.